

PARTHA SINGHA ROY

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Embedded Systems Engineer specializing in C/C++ and Embedded Linux for ARM-based Edge AI systems. Hands-on experience with Yocto, Linux kernel, device drivers, V4L2, DMA-BUF, and Rockchip MPP. Experienced in real-time video pipelines, CNN-based object detection, and AI acceleration across Rockchip and Renesas platforms.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript, CMake, OOP, Bare Metal

Embedded Linux: Linux, Yocto Project, BitBake, U-Boot, Linux Kernel, Device Drivers, Device Tree, IPC, POSIX Threads

AI / Computer Vision: CNN, YOLOv3, YOLOv5, YOLOv8, Object Detection, Edge AI, NPU, DRP-AI, OpenCV

Multimedia: V4L2, DMA-BUF, Rockchip MPP, GStreamer, FFmpeg, H.264, RTP/RTSP

Hardware: Rockchip RK3576, Renesas RZ/V2L, RZ/V2M, NVIDIA Jetson, Raspberry Pi, ESP32/ESP32-C3

Tools: Git, VS Code, ESP-IDF, PlatformIO, Altium Designer, KiCad, LTspice

WORK EXPERIENCE

Embedded Software Engineer

June 2026 - Present

Naveera Tech.

Denver, Colorado, United States · Remote

- Developing an AI-enabled dashcam and collision-detection system in C/C++ on the Rockchip RK3576 platform, leveraging its 6 TOPS NPU for real-time on-device AI inference and object detection.
- Implementing a dual-camera video pipeline using IMX415 and OV5647 sensors with V4L2 capture, DMA-BUF buffer sharing, H.264 encoding, and 60 FPS recording.
- Designing concurrent software components using Inter-Process Communication (IPC) and POSIX threads for camera capture, video encoding, AI processing, and event handling.
- Developing high-performance, low-latency video processing components using V4L2, DMA-BUF, and Rockchip MPP for efficient zero-copy data handling.

Embedded System Engineer

Jul 2024 - June 2026

Aarish Technologies Inc.

New Town, Kolkata

- Built a C++ multi-threaded POSIX application for audio CNN-based Battlefield Surveillance Radar sound classification and implemented logarithmic and linear AGC algorithms.
- Engineered a custom SDR communication system on PlutoPlus for real-time H.264 video and drone telemetry transmission using OFDM-QPSK baseband processing with liquid-dsp; implemented wideband frequency hopping across 900 MHz–5.5 GHz and achieved data rates up to 735 kbps.
- Created a C/C++ smart vision system on Raspberry Pi and ESP32-C3 using YOLOv3, RTP streaming, GPS/PIR/MEMS sensors, and ping-pong frame buffering.
- Developed a modular C++ framework for Renesas RZ/V2L to execute CNN models on the DRP-AI accelerator; built a Yocto-based Embedded Linux platform with kernel customization, DTS, and board-specific layers.
- Integrated IT6161 HDMI-MIPI bridge, IMX219/OV5647 camera sensors, and Wi-Fi drivers in Linux kernel 5.10; led RZ/V2L SBC design and production, reducing BOM cost by 40%.

Software Engineering Intern

Jan 2023 – Jun 2023

FOSSEE, IIT BOMBAY

Remote

- Enhanced eSim software for Ubuntu 22.04, including upgrading the integrated KiCad from v5 to v6.

Hardware Design Intern

Feb 2023 – April 2023

VLSI SYSTEM DESIGN, IIT BOMBAY

Remote

- Designed custom CMOS designs using SKY130 PDK, performing schematic capture (Xschem), layout (Magic/ALIGN), and post-layout characterization (NGSpice); implemented a 1-bit ADC in the 130 nm node.

PROJECTS

DIGITAL SIGNAL PROCESSING IN ESP32 | C/C++, RTOS

Built a dual-core real-time signal-processing system with 16-bit FFT, core-level task segregation, and Wi-Fi data streaming to a web interface.

EDUCATION

Bachelor of Technology (B.Tech.) in Electronics & Communication Engineering

Aug. 2020– July 2024

Kalyani Govt. Engineering College

Kalyani, Nadia, West Bengal